



Swami Vivekananda Yoga Anusandhāna Samsthāna (S-VYASA)
(Deemed to be University)

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Examination	Internal Assessment - 1	Academic year	2025- 2026
Programme	B. Tech	Specialization	All
Semester	01	Date of Examination	24-09-2025
Course Code	BTNX111	Session	AN
Course Name	Engineering Mathematics 1		

Maximum Duration: 1 Hour

Maximum Marks: 30

Part – A

Answer all the questions

5 Q X 3 M =15

Q. No.	Questions	CO	RBTL	MARKS
1.	Evaluate $(2F)_{16} + (A1)_{16}$	2	3	3
2.	What are the various methods of representing two's complement?	1	1	3
3.	If $A=9$ and $B= 11$ find $(A \text{ NAND } B)$.	1	3	3
4.	Find GCD (18,30) using the Euclidean Algorithm.	2	3	3
5	State and explain the Shift operators with examples.	1	2	3

PART-B

Answer all the questions

3 Q X 5 M =15

Q. No	Questions	CO	RBTL	MARKS
6a.	For $A = 13$ and $B = 11$ Does $\sim(A \text{ AND } B)$ equal $(A \text{ NAND } B)$?	1	3	5

OR

6b.	If $A=25$ and $B=14$ find $C = (A \text{ XOR } B) \text{ AND } (A \text{ OR } B)$	1	3	5
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7a.	State modulo properties and explain each with examples.	2	3	5
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OR

7b.	Find the Multiplicative Inverse of 9 mod 13 using Extended Euclidean Algorithm.	2	3	5
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8a.	If the LCM of two numbers is 495 and their GCD is 5, and the sum of the two numbers is 100, find the numbers.	2	3	5
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OR

8b.	Given that $a = 24$ and b is eight more than 'a'. Find GCD, LCM and then verify $\text{GCD} \times \text{LCM} = a \times b$.	2	3	5
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